AS/400 Display File vs React Frontend

Inventory Management — Screen Comparison Report

Project: TRTCT004 — Inventory & Payroll Management System

Application: Inventory Management (trtct004-inventory)

Source System: IBM AS/400 RPG — Programs INVMNUR, INVLSTR, INVDTLR, INVRPTR

Target System: Spring Boot 3.3.4 + React 18 + TypeScript + Material UI 5

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1. Screen-Level Mapping

#	AS/400 DISPLAY FILE (DSPF)	LEGACY RPG PROGRAM	REACT FRONTEND COMPONENT	ROUTE
1	SCR-001 — Inventory Main Menu	INVMNUR (PGM-001)	InventoryMenu.tsx	/inventory/menu
2	SCR-002 — Inventory Item List	INVLSTR (PGM-002)	InventoryList.tsx	/inventory/list
3	SCR-003 — Item Detail (Add)	INVDTLR (PGM-003)	InventoryItemDetail.tsx	/inventory/items/new?mode=A
4	SCR-004 — Item Detail (Edit/Display)	INVDTLR (PGM-003)	InventoryItemDetail.tsx	/inventory/items/:id?mode=E or D
5	SCR-005 — Exit Confirmation	INVMNUR (PGM-001)	ExitConfirmation.tsx	/exit
6	SCR-006 — Reorder Report	INVRPTR (PGM-004)	ReorderReport.tsx	/inventory/report/reorder

2. SCR-001: Main Menu — Field-Level Comparison

FIELD ID	AS/400 DSPF FIELD	TYPE	REACT EQUIVALENT	UI CHANGE
F-001-01	Screen Title (constant)	Output	`<Typography variant="h4">` in hero section	Static text → branded hero card with icon
F-001-02	System Date (MNSDAT)	Output	Not displayed	**REMOVED** Browser provides date natively
F-001-03	System Time (MNSTIM)	Output	Not displayed	**REMOVED** Not needed in web UI
F-001-04	Option 1 label	Output	`NAV_CARDS[0]` = "Browse Inventory"	Text label → Material UI Card with icon + description
F-001-05	Option 2 label	Output	`NAV_CARDS[1]` = "Add New Item"	Card with gradient icon
F-001-06	Option 3 label	Output	`NAV_CARDS[2]` = "Reorder Report"	Card with gradient icon
F-001-07	Option 9 label (Exit)	Output	Not on menu screen	Exit moved to sidebar/AppLayout navigation
F-001-08	Error Message (MNERR)	Output	Not applicable	Replaced by toast/Snackbar notifications
F-001-09	Selection Entry (MNSEL)	Input, 1-char	`onClick` on each `<CardActionArea>`	**KEY CHANGE** Typed numeric input → clickable navigation cards with React Router

Interaction Paradigm Shift: The legacy green-screen menu required typing a single digit (1/2/3/9) + Enter. The React version uses visual navigation cards with hover animations and click-to-navigate via React Router v6. The function key paradigm (F3=Exit) is replaced by browser back navigation and an explicit exit route.

3. SCR-002: Inventory Item List — Field-Level Comparison

FIELD ID	AS/400 DSPF FIELD	TYPE	REACT EQUIVALENT	UI CHANGE
F-002-01	Screen Title	Output	`AppLayout title="Inventory Item List"`	Rendered in reusable layout header
F-002-02	Op Column (LSOPT)	Input, 1-char/row	3 `<IconButton>` per row: Edit, View, Delete	KEY CHANGE "2=Edit 4=Delete 5=Display" codes → individual icon buttons with tooltips
F-002-03	Item ID (LSITID)	Output	`<TableCell>` monospace, indigo color	Same data, enhanced with color coding
F-002-04	Item Name (LSITNM)	Output	`<TableCell>` bold font	Same data
F-002-05	Category Code (LSCTGY)	Output	`<Chip>` badge, monospace	Displayed as pill/badge instead of plain text
F-002-06	Qty On Hand (LSQTOH)	Output, numeric	`<TableCell>` right-aligned, `.toLocaleString()`	ENHANCED Thousands separators
F-002-07	Unit Price (LSUNPR)	Output, numeric	`<TableCell>` with $ prefix, green color	ENHANCED Currency formatting + color
F-002-08	Status (LSSTAT)	Output, 1-char	Color-coded `<Chip>`: ACTIVE / INACTIVE / PENDING_DELETE / DELETED	ENHANCED Single char → full-word color-coded badges
F-002-09	Error Line (LSERR)	Output	`<Snackbar><Alert>` toast notification	Inline error → floating notification
F-002-10	Record Count / "Bottom"	Output	Footer bar: `"{n} records found"`	Dynamic record count in table footer
F-002-11	Category Filter (SRCTGY)	Input, 6-char	`<TextField label="Category">` with search icon	Text input with adornment
F-002-12	Name Filter (SRNAME)	Input, 15-char	`<TextField label="Item Name">` with search icon	Text input with adornment
—	F3=Exit / F5=Refresh / F6=Add	Function Keys	Refresh IconButton + Add New Item Button	KEY CHANGE Function keys → visible buttons
—	SFL/SFLCTL records	Subfile	`<Table>` with `.map()` rendering	Subfile format → MUI Table with array map

Interaction Paradigm Shift: The AS/400 subfile list (SFL/SFLCTL) used a fixed-row output with a single "Op" input column where users typed action codes (2/4/5) next to each row and pressed Enter to process all at once. The React version provides individual action buttons per row with immediate async execution. Filtering is live (triggers on change

> with `useCallback`), replacing the Enter-to-submit filter model.

4. SCR-003/004: Item Detail — Field-Level Comparison

FIELD ID	AS/400 DSPF FIELD	TYPE	REACT EQUIVALENT	UI CHANGE
F-003-01	Screen Title	Output	`AppLayout title` = dynamic per mode	Mode-aware title: "Add New Item" / "Edit Item — X" / "View Item — X"
F-003-02	Mode Indicator (DTMODE)	Output	`<Chip>` badge with color per mode	Mode badge in header: Add=primary, Edit=secondary, View=default
F-003-03	Error Message Area (DTERR)	Output	`<Snackbar>` + per-field `helperText`	**ENHANCED** Single error line → field-level inline errors + toast
F-003-04	Save Success (DTSAV)	Output, indicator	`<Alert severity="success">` via Snackbar	Toast notification
F-003-05	Item ID (DTITID)	Input/Output, 8-char	`<TextField>` via react-hook-form Controller	Disabled in Edit/Display mode via `MODE_CONFIG`
F-003-06	Item Name (DTITNM)	Input, 30-char	`<TextField>` + Zod `.min(1).max(30)`	**ENHANCED** Client-side validation added
F-003-07	Description (DTITDS)	Input, 50-char	`<TextField>` with `.max(50)`	Same
F-003-08	Category Code (DTCTGY)	Input, 6-char	`<TextField>` monospace, `maxLength: 6`	Same
F-003-09	Category Name (DTCTNM)	Output (enriched)	Read-only `<TextField>` from `serverData.categoryName`	Enriched from adapter lookup, display-only
F-003-10	Supplier Code (DTSUPL)	Input, 6-char	`<TextField>` monospace	Same
F-003-11	Supplier Name (DTSUNM)	Output (enriched)	Read-only `<TextField>` from `serverData.supplierName`	Enriched from adapter lookup
F-003-12	Warehouse Code (DTWHSE)	Input, 4-char	`<TextField>` monospace	Same
F-003-13	Warehouse Name (DTWHNM)	Output (enriched)	Read-only `<TextField>` from `serverData.warehouseName`	Enriched from adapter lookup
F-003-14	Qty On Hand (DTQTOH)	Input, numeric	`<TextField type="number">` with `min: 0`	Numeric input with validation
F-003-15	Qty Allocated (DTQTAL)	Output (read-only)	Read-only `<TextField>` with info tooltip	System-calculated, not user-editable
F-003-16	Qty On Order (DTQTOR)	Output (read-only)	Read-only `<TextField>` with info tooltip	System-calculated

FIELD ID	AS/400 DSPF FIELD	TYPE	REACT EQUIVALENT	UI CHANGE
F-003-17	Reorder Point (DTROPL)	Input, numeric	`<TextField type="number">`	Same
F-003-18	Reorder Quantity (DTROQT)	Input, numeric	Present in DTO, not in current form layout	Field available but not rendered on screen
F-003-19	Unit Cost (DTUNCS)	Input, decimal 9,2	`<TextField type="number" step="0.01">`	Decimal step input
F-003-20	Unit Price (DTUNPR)	Input, decimal 9,2	`<TextField>` + Zod `refine(price ≥ cost)`	ENHANCED Client-side cross-field validation (BR-045)
F-003-21	Gross Margin % (DTMRGN)	Output, derived ALG-001	Computed live: `((price-cost)/price)*100` with color	ENHANCED Live-updating + color indicator (green ≥20%, yellow ≥0%, red <0%)
F-003-22	Status (DTSTAT)	Output	Read-only `<TextField>`	Same
—	F3=Cancel / F12=Return	Function Keys	Cancel / Back to List Button	Explicit buttons replace F-keys
—	Enter=Save	Function Key	Save Button with loading spinner	Click-to-save with `isSubmitting` state
—	—	—	Edit This Item Button (View mode only)	NEW Inline mode switch from Display → Edit

Interaction Paradigm Shift: The legacy display file used a single record format with conditional field attributes (INPUT/OUTPUT based on mode indicator). The React version uses Zod schema validation at the form level (client-side), `react-hook-form` for state management, and the same component renders all three modes (Add/Edit/Display) with field-level disable states controlled by `MODE_CONFIG`. Gross margin is computed live as the user types (not on Enter).

5. SCR-005: Exit Confirmation — Field-Level Comparison

FIELD ID	AS/400 DSPF FIELD	TYPE	REACT EQUIVALENT	UI CHANGE
F-005-01	Screen Title	Output	`<Typography variant="h5">` "Exit Application?"	Centered modal-style card
F-005-02	Exit Prompt (BR-005)	Output	`<Typography>` description text	Descriptive warning text with icon
—	Enter=Confirm / F12=Cancel	Function Keys	Confirm Exit + Cancel Buttons	KEY CHANGE Function keys → explicit styled action buttons

Interaction Paradigm Shift: The legacy exit confirmation was a simple Y/N prompt on the green screen. The React version renders a full-screen glassmorphic modal with animated background, a Logout icon, and explicit button actions with hover effects.

6. SCR-006: Reorder Report — Field-Level Comparison

FIELD ID	AS/400 DSPF FIELD	TYPE	REACT EQUIVALENT	UI CHANGE
F-006-01	Screen Title	Output	`AppLayout title="Reorder Report"`	Layout header with subtitle
F-006-02	Item ID (RPITID)	Output	`<TableCell>` monospace, indigo	Same
F-006-03	Item Name (RPITNM)	Output	`<TableCell>` bold	Same
F-006-04	Category (RPCTGY)	Output	`<Chip>` badge	Pill badge style
F-006-05	Qty On Hand (RPQTOH)	Output, numeric	`<TableCell>` right-aligned, locale formatted	Thousands separators
F-006-06	Reorder Point (RPROPL)	Output, numeric	`<TableCell>` right-aligned	Same
F-006-07	Shortage Qty (RPSHRT)	Output, derived ALG-002	`<TableCell>` red bold, right-aligned	ENHANCED Color-coded red for urgency
F-006-08	Supplier Code (RPSUPL)	Output	`<TableCell>` monospace	Same
F-006-09	Repl. Value (RPRVAL)	Output, derived ALG-003	`<TableCell>` with $ + locale currency	ENHANCED Currency formatting
F-006-10	Summary Totals (RPTCNT, RPTVAL)	Output, footer	Two `<Card>` stat summary cards with gradient icons	ENHANCED Summary line → visual KPI cards

FIELD ID	AS/400 DSPF FIELD	TYPE	REACT EQUIVALENT	UI CHANGE
F-006-11	Warehouse Filter (RPWHSE)	Input, 4-char	`<TextField>` with placeholder "All warehouses"	Descriptive placeholder
F-006-12	Category Filter (RPCTGY)	Input, 6-char	`<TextField>` with placeholder "All categories"	Descriptive placeholder
—	F5=Refresh / F6=Print	Function Keys	Refresh + Print Report Buttons	Explicit buttons
—	—	—	Urgency column: Critical / High / Low `<Chip>`	NEW Urgency indicator derived from shortage ratio
—	—	—	Client-side sort by shortage descending (DT-008)	NEW Items sorted by urgency in frontend

Interaction Paradigm Shift: The legacy INVRPTR report was a static print-oriented subfile. The React version adds interactive filter controls, live refresh, visual KPI summary cards, an urgency classification column derived from shortage-to-reorder-point ratio, and client-side sorting by severity — none of which existed in the original display file.

7. Cross-Cutting Differences Summary

ASPECT	AS/400 DISPLAY FILE (DSPF)	REACT FRONTEND
Rendering	Server-rendered 5250 terminal stream, 24×80 fixed character grid	Client-side SPA, responsive Material UI grid with CSS flexbox/grid
Navigation	Function keys (F3, F5, F6, F12) + typed selection codes	React Router v6 + clickable buttons / navigation cards
Input Model	Typed codes per field, Enter to submit all fields at once	Individual form controls, click/type with live validation
Validation	Server-side only (RPG program logic after Enter)	Dual: client-side Zod schema validation + server-side Spring validator
Error Display	Single error message line at fixed screen row position	Per-field inline helperText errors + floating Snackbar toasts
List Actions	"Op" column: type 2/4/5 next to rows, Enter to batch-process	Per-row icon buttons (Edit/View/Delete) with immediate async execution
Computed Fields	Calculated server-side, refreshed on Enter	Live-updating on keystroke (e.g., ALG-001 gross margin)
Styling	Green-on-black monochrome, fixed-width font	Dark glassmorphic theme, gradient icons, animations, color-coded status chips
State Management	Stateful server session (RPG program call stack)	Stateless REST API + client-side React state (useState / react-hook-form)
Printing	OVRPRTF → spool file → output queue	HTTP POST to print endpoint → backend generates spool
List/Subfile	SFL/SFLCTL record formats with SFLSIZ/SFLPAG paging	`Array.map()` into `<Table>` rows, unlimited scroll
New Features	—	Urgency indicators, live margin color, skeleton loading states, responsive layout, inline mode-switch button, KPI summary cards

8. Business Rule Preservation Matrix

RULE ID	DESCRIPTION	LEGACY LOCATION	MODERN LOCATION	STATUS
BR-002–007	Menu option routing & validation (DT-001)	INVMNUR RPG + SCR-001 DSPF	InventoryMenu.tsx + InventoryMenuController.java	✓ Preserved

RULE ID	DESCRIPTION	LEGACY LOCATION	MODERN LOCATION	STATUS
BR-008–022	List browse, filter, action dispatch (DT-002/003)	INVLSTR RPG + SCR-002 SFL	InventoryList.tsx + InventoryListController.java	✓ Preserved
BR-036–048	Field validation: ID, name, codes, price≥cost (DT-005)	INVDTLR RPG logic	InventoryItemValidator.java + Zod schema in InventoryItemDetail.tsx	✓ Preserved (dual layer)
ALG-001	Gross margin: (price−cost)/price×100	INVDTLR RPG CALC	InventoryItemService.calculateMargin() + client-side in InventoryItemDetail.tsx	✓ Preserved
ALG-002	Shortage: max(0, reorderPoint−qtyOnHand)	INVRPTR RPG CALC	ReorderReportService.calculateShortage()	✓ Preserved
ALG-003	Replenishment value: shortage × unitCost	INVRPTR RPG CALC	ReorderReportService.calculateReplenishmentValue()	✓ Preserved
BR-049–051	Create item: init alloc=0, onOrder=0, status=ACTIVE	INVDTLR RPG WRITE	InventoryItemService.createItem()	✓ Preserved
BR-016–020	Soft delete: block if allocated, ACTIVE→INACTIVE	INVLSTR RPG	InventoryItemService.softDeleteItem() + STM-001	✓ Preserved
STM-001	State machine: T-001/T-002/T-003 transitions	Implicit in RPG program flow	InventoryItemStateMachine.java + InventoryItemEntity methods	✓ Preserved (formalized)
BR-055–070	Reorder eligibility, aggregation, 200-item cap	INVRPTR RPG	ReorderReportService.aggregateReorderData()	✓ Preserved